AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 23 · STUDENT LESSON

Fair for Whom?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Count every assessed case. Inspect every relevant condition.

Compare shared tests, trace weighted totals and audit missing outcomes before deciding what a performance claim supports.

Keep one habit: Read the denominator → compare conditions → state the trade-off and evidence limit.

MAIN LESSON · SECTION 1

Who does the headline score leave out?

> Read the denominator before comparing the result.

Start with a performance claim

A system can lead overall and still perform worse in one condition. This week, count correct and assessed cases, inspect each condition, and explain what these small tests leave unknown.

Recall selected and assessed cases

Eight cases were selected, but one has no assessed outcome. How many have assessed outcomes, and what must a performance report say about the remaining case?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Your learning goals

- I can compute a correct-out-of-assessed fraction and explain its **denominator**.
- I can compare systems on shared cases while checking performance within conditions.
- I can combine counts correctly and keep missing outcomes separate from assessed errors.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

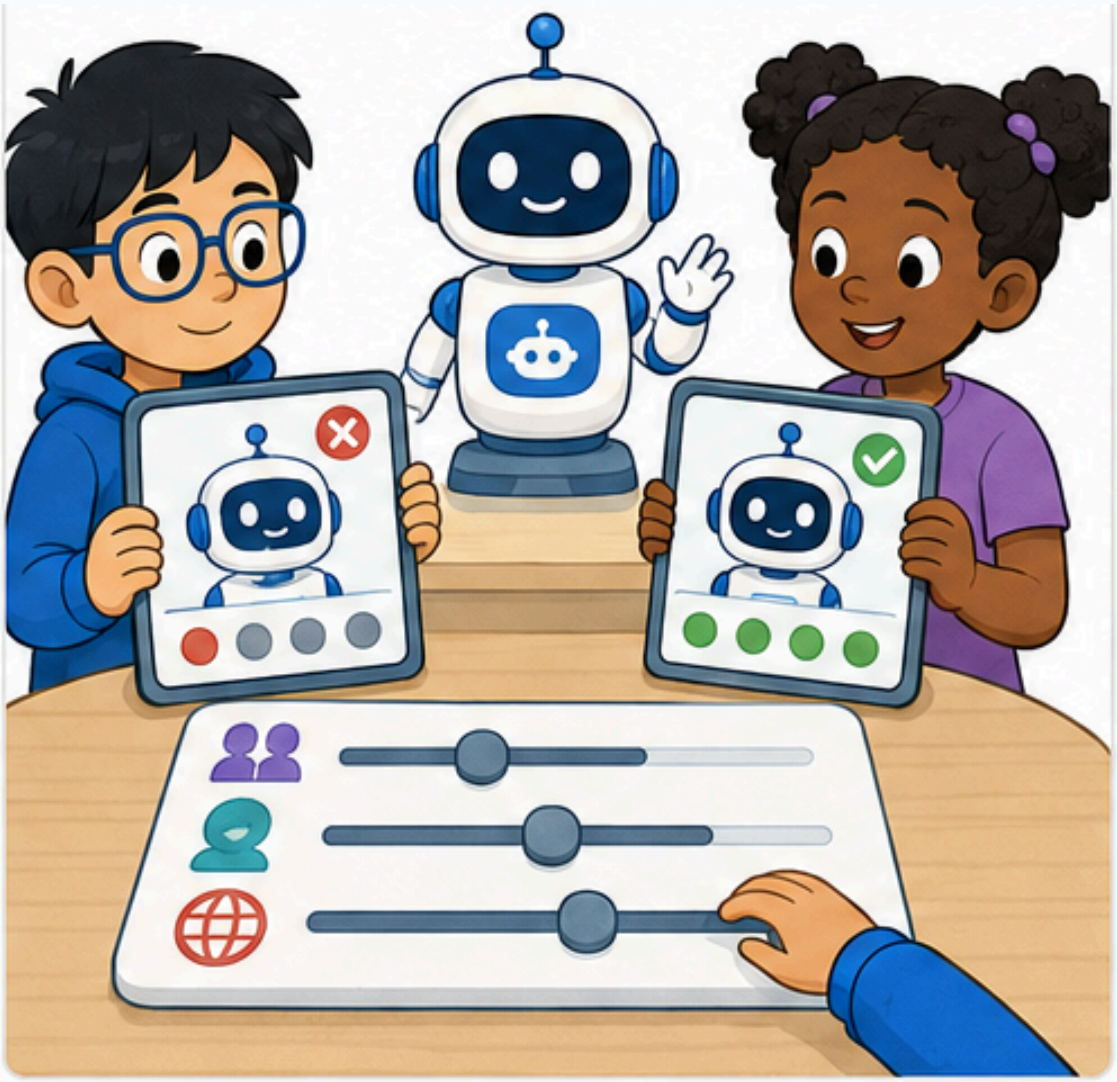
Remember before reading

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

Fairness requires checking outcomes for different users and situations.



LEARNING TARGET System A succeeds on 6 of 8 quiet tests and 3 of 4 noisy tests. Is its noisy success rate lower? Show fractions and percentages. What can these small tests not establish?

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can compare group fractions and combine unequal-sized groups correctly.

MAIN LESSON · SECTION 2

Meet the shared 12-card test

> Keep versions, reference labels and assessed cases fixed for a meaningful comparison.

Define the performance fraction

For this week, **accuracy** means the number of correctly labelled cases divided by the number of assessed cases.

Every assessed case has a system label and a checked reference label, so its result can be judged correct or incorrect. State the system version, cases and scoring rule with the fraction.

Cases without an assessable outcome are reported separately.

Read the complete A/B test

Two fixed shape-recognition versions, A and B, label the same 12 previously unseen picture cards. Eight cards are bright and four are dim; each has a checked circle or square reference label.

Both versions return labels for all 12. A is correct on 8 bright and 1 dim card. B is correct on 6 bright and 2 dim cards.

Calculate A and B from the counts

A therefore has 9 correct out of 12 assessed, written 9/12. Dividing **numerator** and **denominator** by 3 gives 3/4. Three quarters equals 75 out of 100, or 75%. B has 8/12, equal to 2/3.

For this comparison keep B as an exact fraction; a rounded percentage is unnecessary.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A voice game may work well overall yet miss more commands from quiet speakers than loud speakers.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

MAIN LESSON · SECTION 3

Compare proportions without inventing evidence

> Equivalent fractions change the notation, not the number of tested cards.

Inspect the condition-by-condition counts

Combine correct and assessed counts

CONDITION	VERSION A	VERSION B
Bright: 8 cards	8 correct of 8	6 correct of 8
Dim: 4 cards	1 correct of 4	2 correct of 4
All: 12 cards	9 of 12 = 3 of 4	8 of 12 = 2 of 3

A has the higher overall accuracy on these same 12 cards, while B has the higher accuracy on the 4 dim cards. The overall fraction does not describe every condition equally.

Compare totals on the shared denominator

With the same **denominator** 12, compare the **numerator**s: 9 is greater than 8. A is correct on one more of these cards.

This establishes the ordering on this test under this scoring rule; it does not prove that A will always outperform B on future cards.

Explain equivalent fractions

- To compare fractions with different **denominator**s, rewrite them as **equivalent fractions** with a common **denominator**. For example, 3/4 equals 9/12 and 2/3 equals 8/12.

  Changing the written **denominator** does not create new test cases: it only helps compare the proportions.

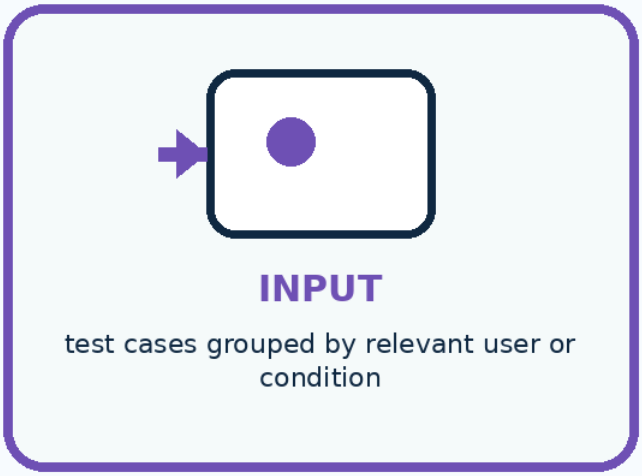
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

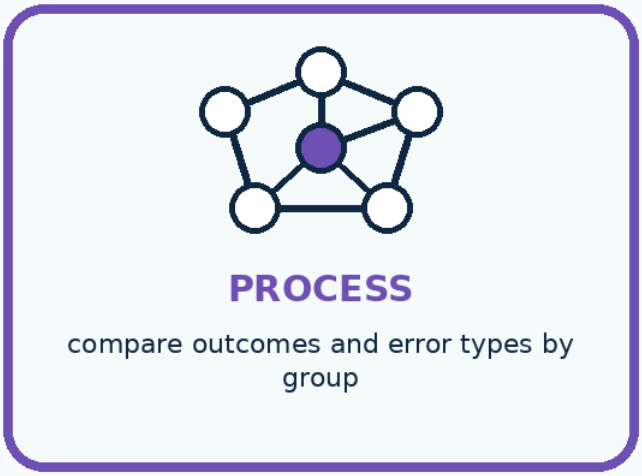
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VISUAL **MECHANISM**

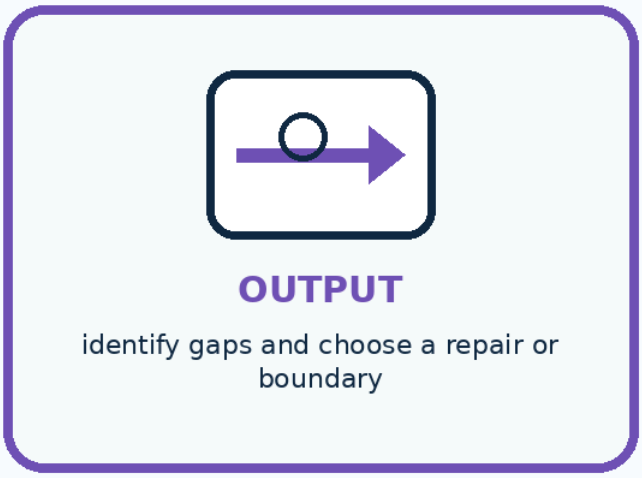
Input → Process → Output

Follow the information instead of imagining magic inside the machine.





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INPUT test cases grouped by relevant user or condition

PROCESS compare outcomes and error types by group

OUTPUT identify gaps and choose a repair or boundary

CORE IDEA **Fairness** requires checking outcomes for different users and situations.

UNDER THE HOOD **Fairness** is context-specific and cannot be established by overall **accuracy** alone. Compare performance, **error cost**s, access, and remedy across relevant groups while avoiding stereotypes and protecting privacy.

Draw a fresh example with all three stages:

A closer explanation

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare group performance and possible remedies

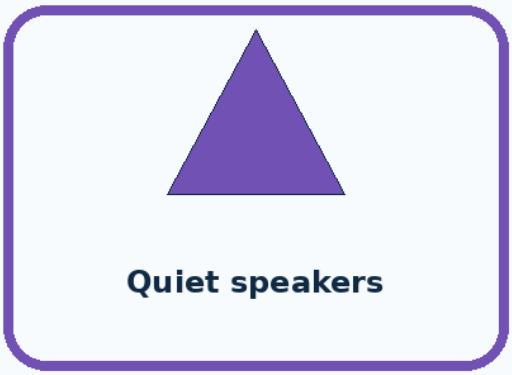
> Inspect the outcomes within conditions before choosing a response.

Pictures paired with their evidence



The case to investigate

CASE 1 — Loud speakers Clue: 9 of 10 commands work. Meaning: high success.



The case to investigate

CASE 2 — Quiet speakers Clue: 5 of 10 commands work. Meaning: performance gap.



The case to investigate

CASE 3 — Overall 14 of 20 Clue: 70 percent total. Meaning: hides difference.



The case to investigate

CASE 4 — Text control Clue: alternate access path. Meaning: possible remedy.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

BOUNDARY **Fairness** work should reduce burdens without turning group labels into stereotypes.

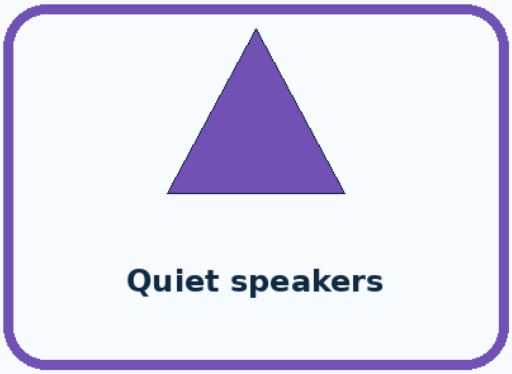
▼ **Earlier wording, alternatives and picture transcripts**

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Loud speakers Clue: 9 of 10 commands work. Meaning: high success.

CASE 2 — Quiet speakers Clue: 5 of 10 commands work. Meaning: performance gap.

CASE 3 — Overall 14 of 20 Clue: 70 percent total. Meaning: hides difference.

CASE 4 — Text control Clue: alternate access path. Meaning: possible remedy.

Change one thing in the pictured case

Change: Give bright and dim groups equal weight as an explicitly different metric.

Prediction: M scores **70%**; N scores about 81.7%.

Reason: This averages group rates and is not the original 16-card agreement rate.

Trace the separate 16-card badge test

> Distinguish per-card accuracy from an explicitly chosen equal-group metric.

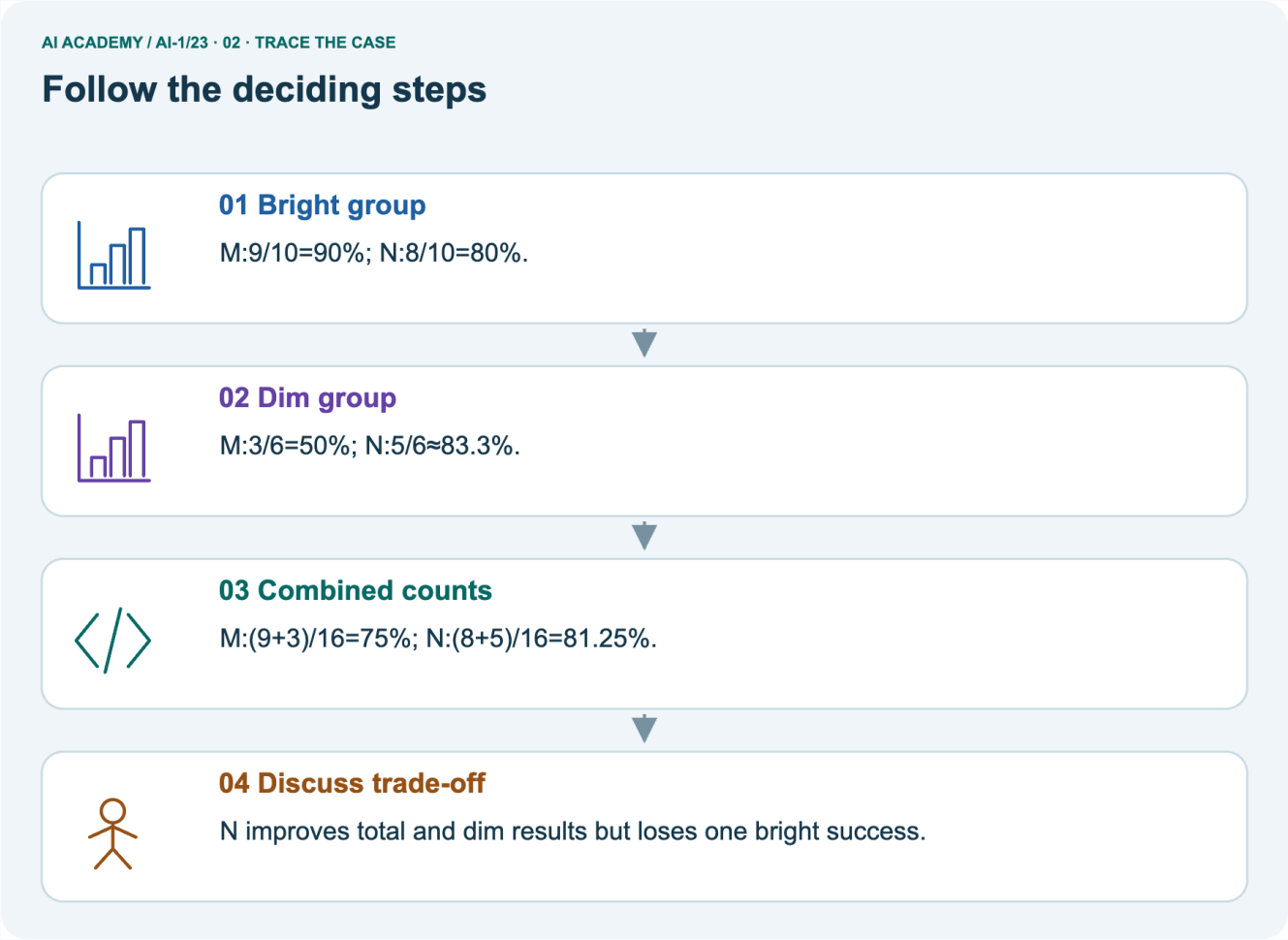
The badge-reader facts

A fictional badge reader is evaluated on bright and dim cards with known reference labels.

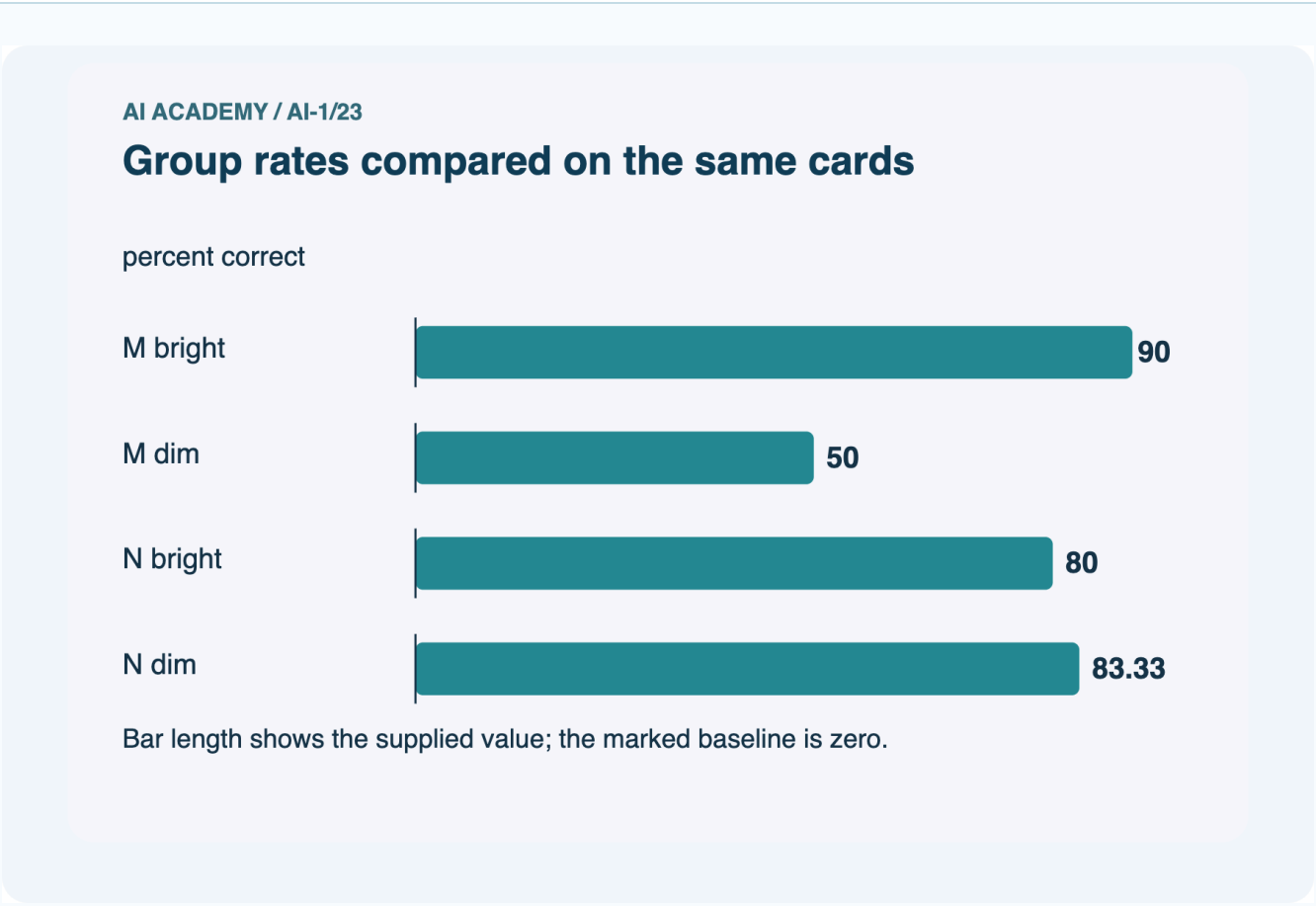
Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.

Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.

Follow the group and total calculations



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Bright group	M:9/10=90%; N:8/10=80%.
2. Dim group	M:3/6=50%; N:5/6≈83.3%.
3. Combined counts	M:(9+3)/16=75%; N:(8+5)/16=81.25%.
4. Discuss trade-off	N improves total and dim results but loses one bright success.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Explain the weighting and trade-off

- Adding correct counts and **denominator**s weights each tested card once. Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

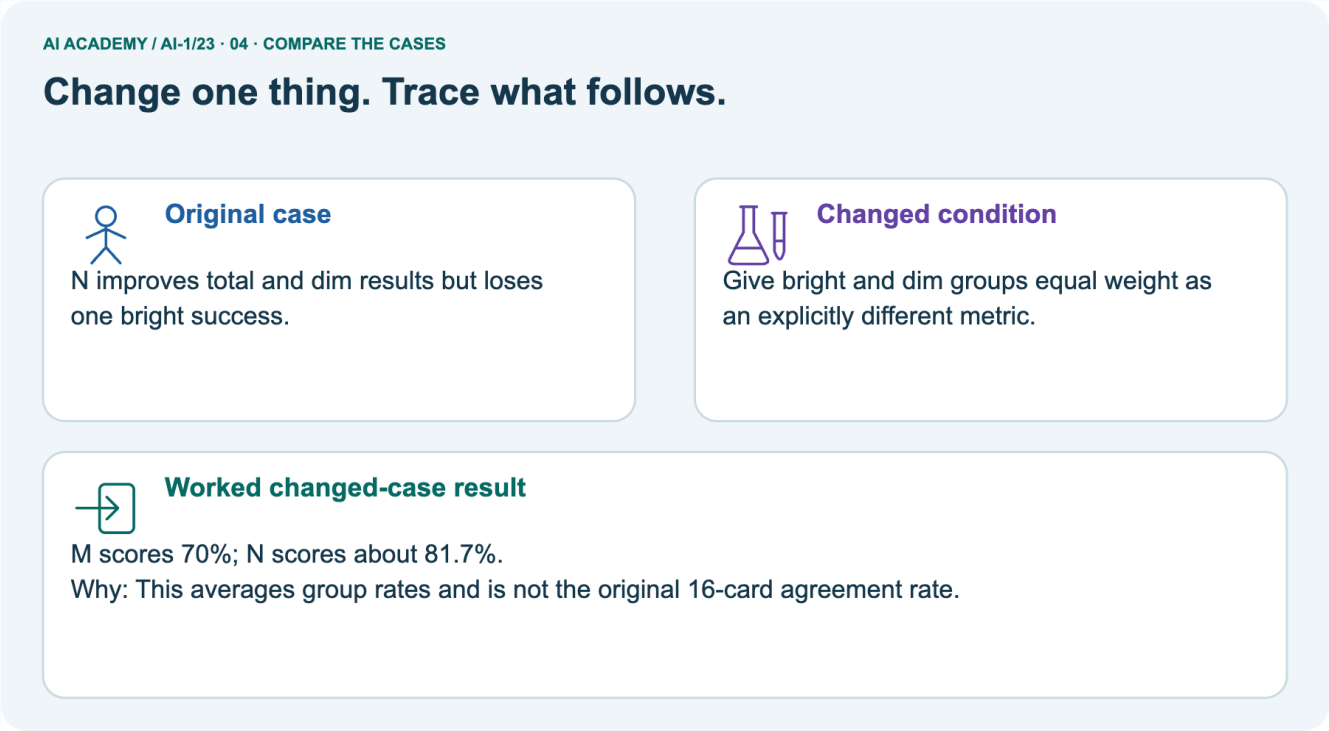
  Decide what matters with the relevant users rather than announcing that one number proves **fairness**.

- Read the labelled picture: Which group loses performance under N, and can total improvement erase that fact?

Predict a different metric

- Before opening the changed-metric picture, predict M and N when bright and dim group rates each receive half the weight. Is this the same question as counting every card once?

▼ Check the changed metric



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the evidence limit

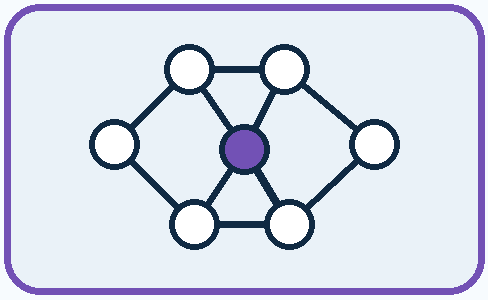
Six dim cards supply limited **evidence**; these descriptive rates do not prove a system is fair for all users.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A voice game may work well overall yet miss more commands from quiet speakers than loud speakers.

STEP 1 — OBSERVE test cases grouped by relevant user or condition

STEP 2 — TRACE compare outcomes and error types by group

STEP 3 — CONCLUDE identify gaps and choose a repair or boundary

MODEL REASONING Report each group's success count and error pattern, not only the total.

Improve microphone/**process** testing with consented varied voices and keep a non-voice control so access does not depend on one **input** mode.

DO NOT OVERCLAIM **Fairness** work should reduce burdens without turning group labels into stereotypes.

What new **evidence** could change this answer?

A closer explanation

Question	Version A	Version B
Bright-card accuracy	8/8 = 1	6/8 = 3/4
Dim-card accuracy	1/4	2/4 = 1/2
Combined accuracy	(8 + 1)/(8 + 4) = 9/12	(6 + 2)/(8 + 4) = 8/12

The bright group has twice as many assessed cards as the dim group. Each individual card counts once in the combined result, so add correct counts and assessed counts.

Do not simply average the two group percentages and call it the all-card **accuracy**: that would give each group equal weight despite their different sizes.

If the intended task especially needs dim-card performance, inspect that result and gather suitable additional **evidence** before choosing. An overall leader can still be weaker in a relevant condition.

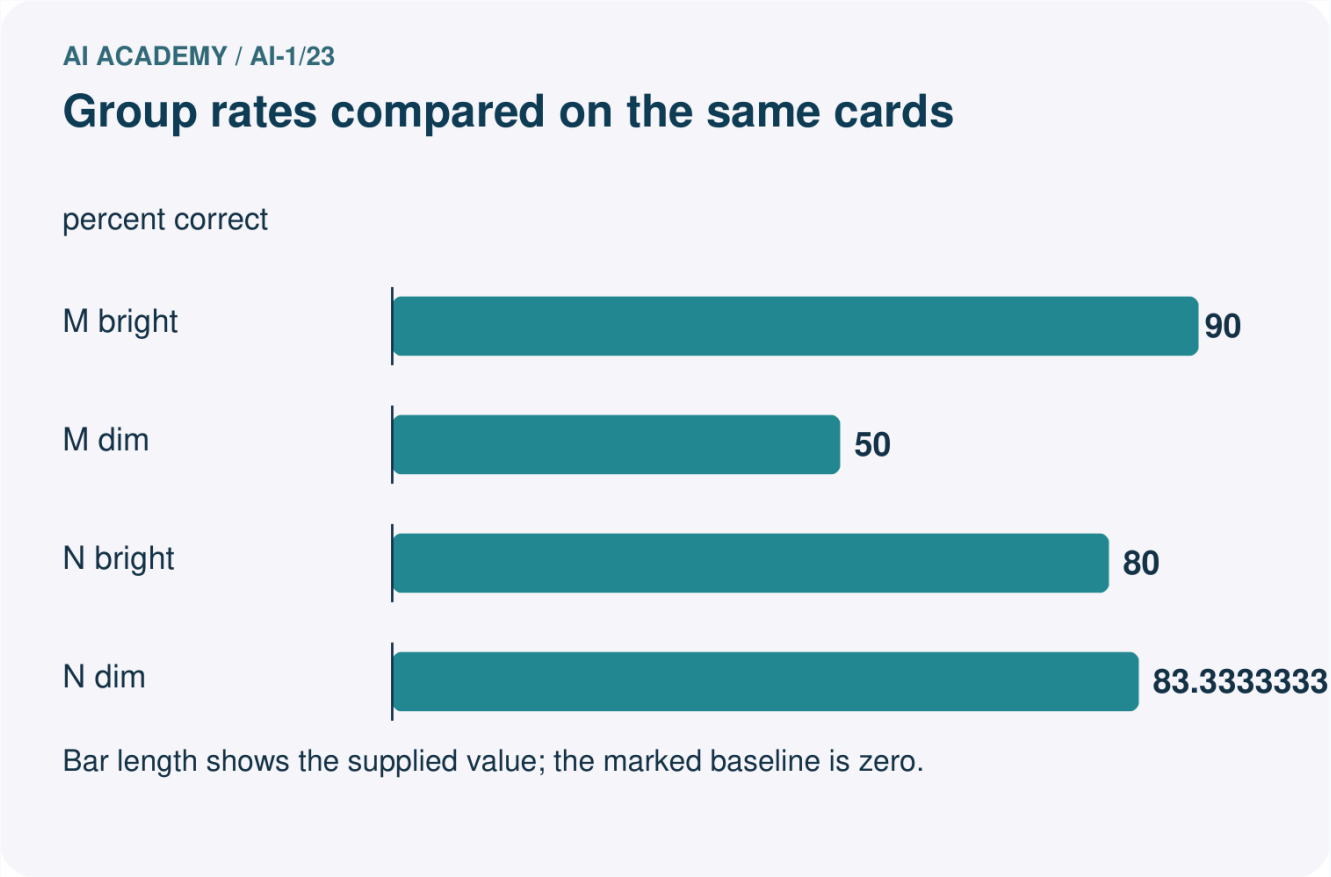
This table alone also does not tell us whether particular mistakes have different consequences. The scoring rule must match the purpose.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Fair for Whom?

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: Give bright and dim groups equal weight as an explicitly different metric.
Predict the result before checking the explanation. What remains the same?

Repair claims that hide the denominator

> A correct percentage still needs counts, coverage and a stated purpose.

Report A with its condition breakdown

A useful performance statement gives the correct count, assessed count, missing count and evaluation conditions. For A: 9/12 correct, no missing outcomes, on the shared test of 8 bright and 4 dim cards.

Follow it with the important contrast: 8/8 bright and 1/4 dim. Percentages are optional shorthand, not replacements for these counts.

Separate proportion from quantity of evidence

A score of 3/4 and a score of 75/100 are equal proportions, but they describe different quantities of **evidence** if they come from different tests. The collection method and case coverage still matter.

Neither fraction is an individual **prediction** confidence or a guarantee about the next case.

Repair both fairness claims

- CLAIM A Good overall **accuracy** proves **fairness**.
- CLAIM B **Fairness** means every group must have identical data or outcomes in every situation.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Loud speakers	9 of 10 commands work	high success
Quiet speakers	5 of 10 commands work	performance gap
Overall 14 of 20	70 percent total	hides difference
Text control	alternate access path	possible remedy

Repair two tempting claims

Repair:

Repair:

A closer explanation

> **A system has 2 correct of 2 in one group and 4 correct of 8 in another. Write the combined correct and assessed counts. Explain why averaging 100% and 50% would answer a different question**

MAIN LESSON · SECTION 8

Practise: compare assessed outcomes honestly

> C has two unassessed outcomes; its headline score cannot complete the shared comparison.

Read C's assessed and missing counts

A third version C is run on the same 12 cards, but two results are lost before they can be judged. Among its 10 assessed outcomes, 9 are correct and 1 is wrong: 9/10 = 90% assessed **accuracy**, with 2 outcomes unassessed.

Do not call this 90% on all 12. Its assessed subset differs from A and B, so the headline fractions do not give the same complete-case comparison.

Choose a defensible next comparison

Recover and assess the two lost results if possible, without changing the versions.

Another clearly labelled comparison can use the exact same cases with known outcomes for every version; disclose which cases it excludes and why. The current totals do not tell us how A performed on C's ten assessed cases.

Never invent that subset score.

Compare totals and conditions in W2

Use workbook W2 for the connected example "Fractions and performance comparisons". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair the denominator mistake

- Use the worked case for Fractions and performance comparisons. Proposed explanation: “Only correct cases belong in the **denominator**”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the badge trade-off

Which group loses performance under N, and can total improvement erase that fact?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Fairness** requires checking outcomes for different users and situations.

Use a fresh example to explain **fairness**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: **Fairness** means every group must have identical data or outcomes in every situation.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

> **A report declares C best because 90% exceeds A's 75%. Write the fraction and missing count for C, explain the comparison problem, and propose an evaluation step that would make the evidence easier to compare**

Practise with less help: Compare totals and conditions

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Investigate equal totals with unequal conditions

> Use the supplied paper case, then distinguish the audit from any actual classroom trial.

The shared recordings and supplied outcomes

Paper investigation: two fixed fictional voice-game versions label the same 12 command recordings with checked reference labels. There are 8 loud and 4 quiet recordings; all outcomes are assessable. Version R is correct on 7 loud and 1 quiet recording. Version S is correct on 5 loud and 3 quiet recordings. Predict the overall and within-condition leaders before calculating.

Predict, calculate and inspect the trade-off

1. Make a table of correct, incorrect and assessed counts for each condition and version. Calculate each overall fraction by adding counts. Explain whether equal totals mean equal service in each condition. Propose one additional test and one alternate control for users whose commands fail. These supplied outcomes are fictional evidence for a paper audit; record “not run” for an actual voice-game test you have not performed.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Record evidence and a practical remedy

Case	Prediction	Observed result	What it means
Normal			
Edge			

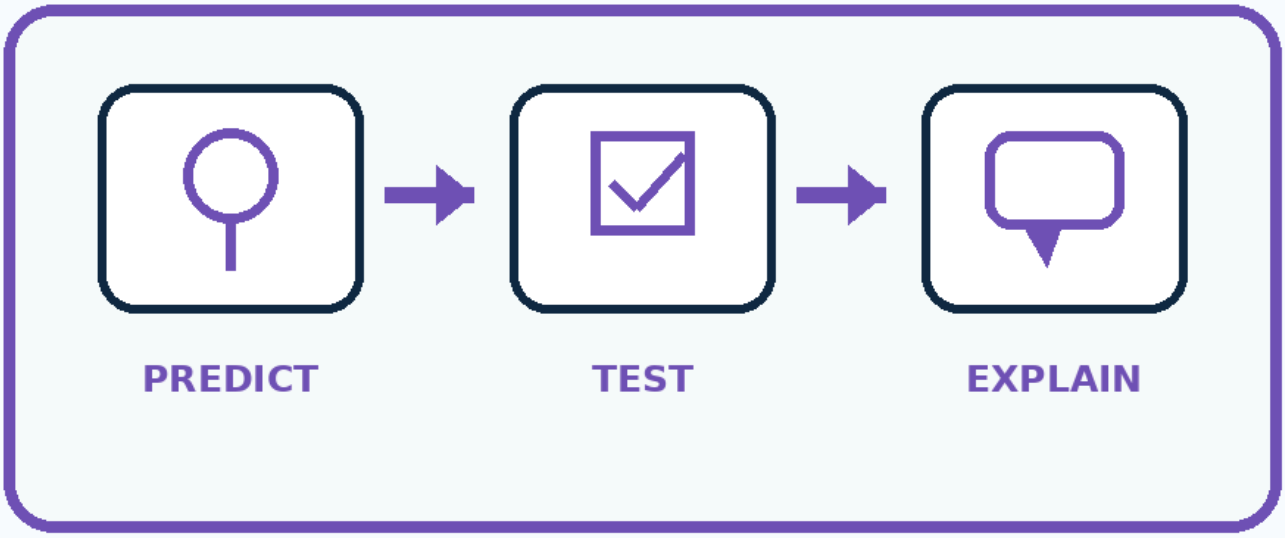
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EXPLORER LAB

Voice Game Fairness Check

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Run a voice-game test for quiet and loud speakers.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare the new shared sound test

> Use the new X/Y outcomes before opening feedback.

The independent task and complete givens

Versions X and Y label the same 20 previously unseen clips: 12 quiet and 8 noisy, with no missing outcomes. X is correct on 10 quiet and 2 noisy clips; Y on 9 quiet and 5 noisy clips.

Calculate each combined fraction and percentage, identify the overall leader and compare the leaders within each group. Explain why an overall result cannot guarantee the next clip's outcome.

Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after your attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Compare a new shared sound test

> **My independent explanation, trace and limits**
>
> _____
>
> _____
>
> _____
>
> _____

Week 23 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A helper succeeds on 4 of 4 clear voices and 1 of 4 quiet voices. Who is less well served? What **evidence** should a repair preserve?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Test the sorter across relevant item conditions and users; never infer sensitive identity to create a **fairness** check.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

System A succeeds on 6 of 8 quiet tests and 3 of 4 noisy tests. Is its noisy success rate lower? Show fractions and percentages. What can these small tests not establish?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Choose sorter checks that reflect its purpose

> Report relevant conditions and the consequences of errors alongside totals.

This week’s project decision

THIS WEEK'S CHECKPOINT Test the sorter across relevant item conditions and users; never infer sensitive identity to create a **fairness** check.

Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision.

Plan evidence, access and uncertainty

- What real classroom need does this serve?
- What information is necessary—and what should stay private?

- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Avoid stereotypes in the remedy

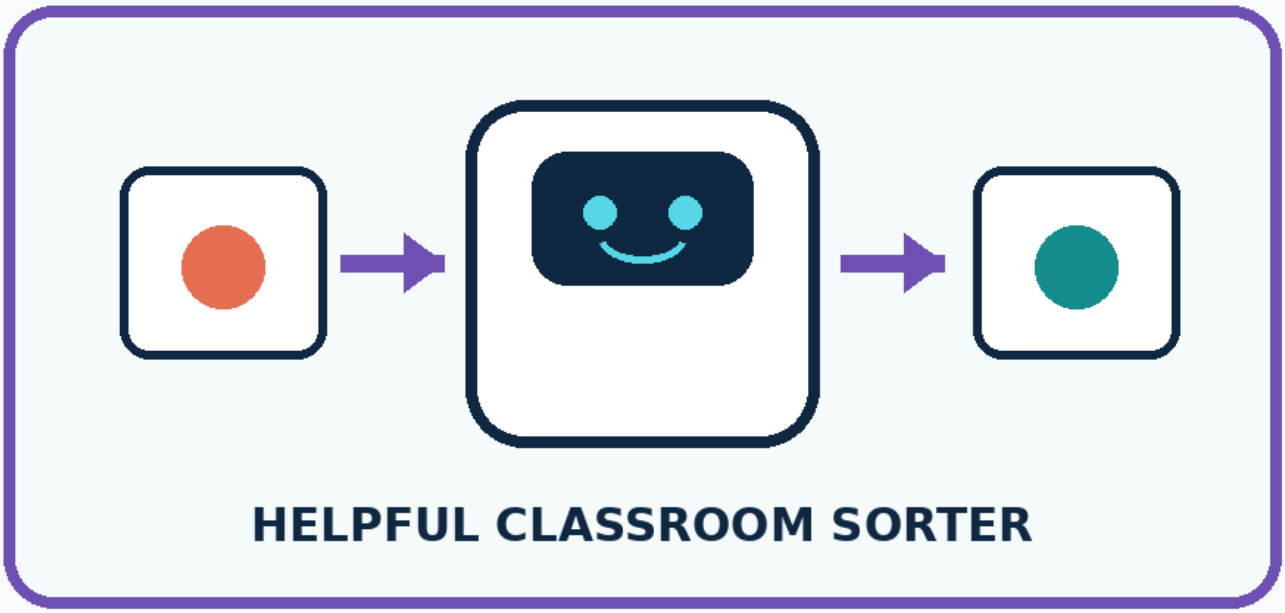
RESPONSIBLE DESIGN **Fairness** work should reduce burdens without turning group labels into stereotypes.

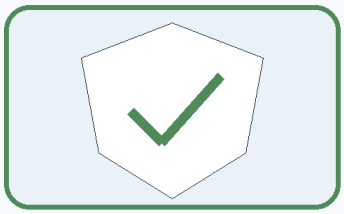
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain the fraction and its limits

> Close your notes and describe both the total and the conditions.

Your exit explanation

- Close your notes. Explain this goal in your own words: Compute a correct-out-of-assessed fraction and explain its **denominator**

My deciding step, evidence and one remaining question

Choose your next check

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
fairness	reasoned treatment of outcomes across people and situations	
subgroup result	performance measured for one relevant group	
gap	difference between group outcomes	
error cost	harm or burden caused by a mistake	

Terms for the connected explanation

Word	Meaning
Numerator	The count above the fraction line; here, assessed correct cases
Denominator	The count below the fraction line; here, all assessed cases
Equivalent fractions	Different written fractions that represent the same proportion

Word	Meaning
Accuracy	The fraction of assessed classification cases labelled correctly under a stated rule
Aggregate	A result formed by combining the counts of several groups

Repair a claim using a word

- CLAIM TO REPAIR Good overall **accuracy** proves **fairness**.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

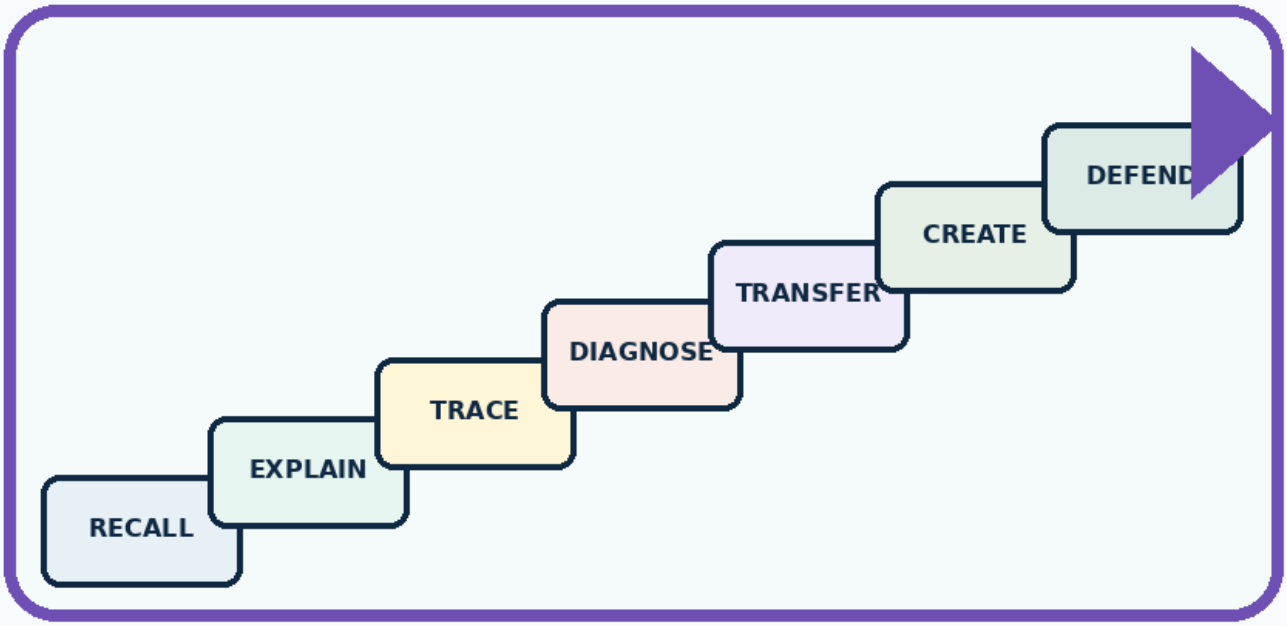
- RECALL Define **fairness**.
- EXPLAIN **Fairness** requires checking outcomes for different users and situations.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Good overall **accuracy** proves **fairness**.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Compare a new shared sound test is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or **fairness**/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

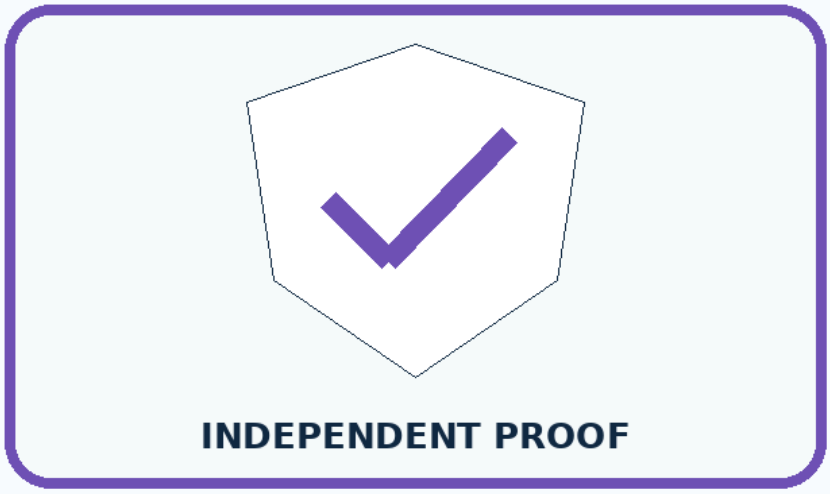
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Compare a new shared sound test is the selected independent case.



EXPLANATION CHECK Compute correct out of assessed, combine counts, compare conditions and state what the test cannot establish.

Independent **evidence**:

SELF-CHECK Report each group's success count and error pattern, not only the total. Improve microphone/**process** testing with consented varied voices and keep a non-voice control so access does not depend on one **input** mode.

NEXT BRIDGE Carry today's **evidence** habit into: Privacy by Choosing Less.

▼ **Extra practice: Compare success under two conditions**

REFERENCE · SECTION 13

Extra practice: Compare success under two conditions

> Try the question before opening its worked explanation.

Read the supplied case and question

A voice system succeeds on 8/10 quiet and 4/10 noisy recordings.

Compare and propose the next investigation.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

The rates differ. Test more varied recordings under controlled conditions and inspect other factors before claiming a cause.

Evidence or stage	Value or result
Quiet	8/10
Noisy	4/10

▼ Earlier wording, alternatives and picture transcripts

Week 23 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Compare success under two conditions

A voice system succeeds on 8/10 quiet and 4/10 noisy recordings.

Compare and propose the next investigation.

The rates differ. Test more varied recordings under controlled conditions and inspect other factors before claiming a cause.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Compute a correct-out-of-assessed fraction and explain its denominator	Explain the deciding step, show a trace or test, and state one limit.
Compare systems on shared cases while checking performance within conditions	Explain the deciding step, show a trace or test, and state one limit.
Combine counts correctly and keep missing outcomes separate from assessed errors	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional badge reader is evaluated on bright and dim cards with known reference labels.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional badge reader is evaluated on bright and dim cards with known reference labels.
- Which labelled step matches this detail? N improves total and dim results but loses one bright success.
- 1. Dim group
- 2. Combined counts
- 3. Bright group
- 4. Discuss trade-off
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Discuss trade-off N improves total and dim results but loses one bright success. Adding correct counts and **denominator**s weights each tested card once.

Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

Decide what matters with the relevant users rather than announcing that one number proves **fairness**.

Changed-case question

- Changed-case practice: Give bright and dim groups equal weight as an explicitly different metric. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. M scores **70%**; N scores about 81.7%.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: M scores **70%**; N scores about 81.7%. This averages group rates and is not the original 16-card agreement rate.

Repair and save your work

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap?

Point to the target groups.. Return to this week and explain what you changed.

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone’s preferences?.

Return to this week and explain what you changed.

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

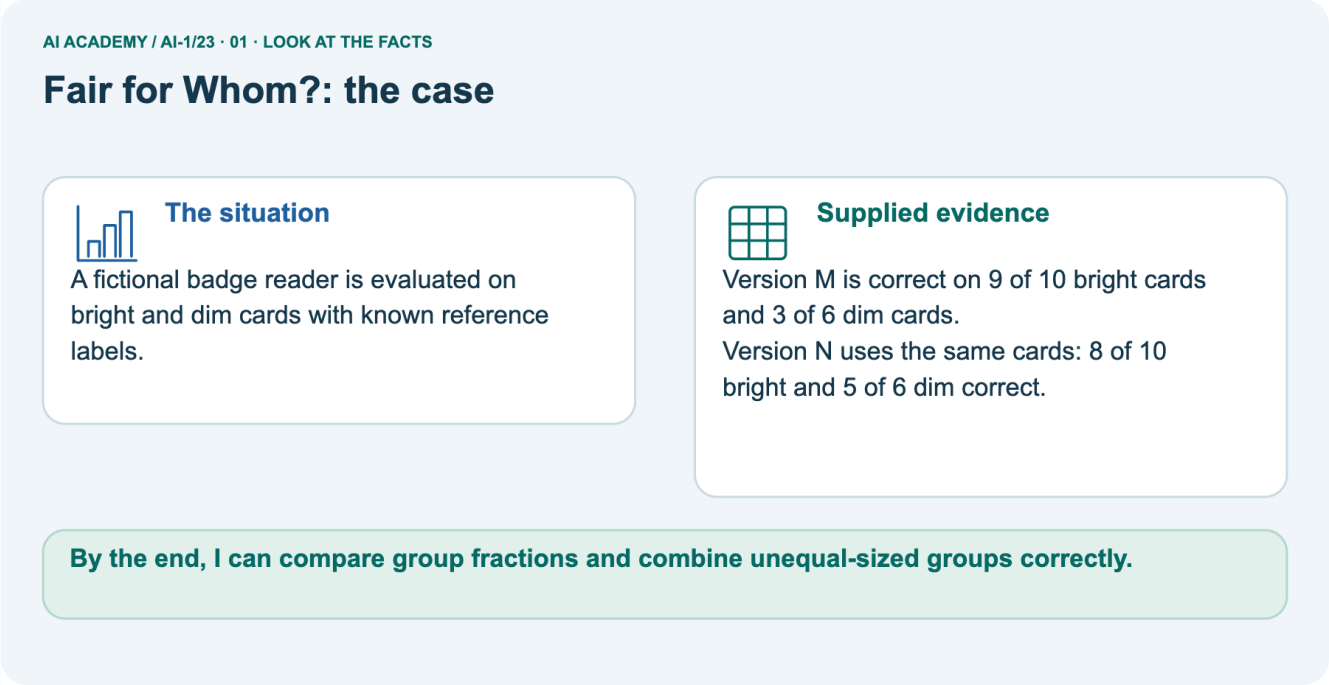
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Fair for Whom?: the case



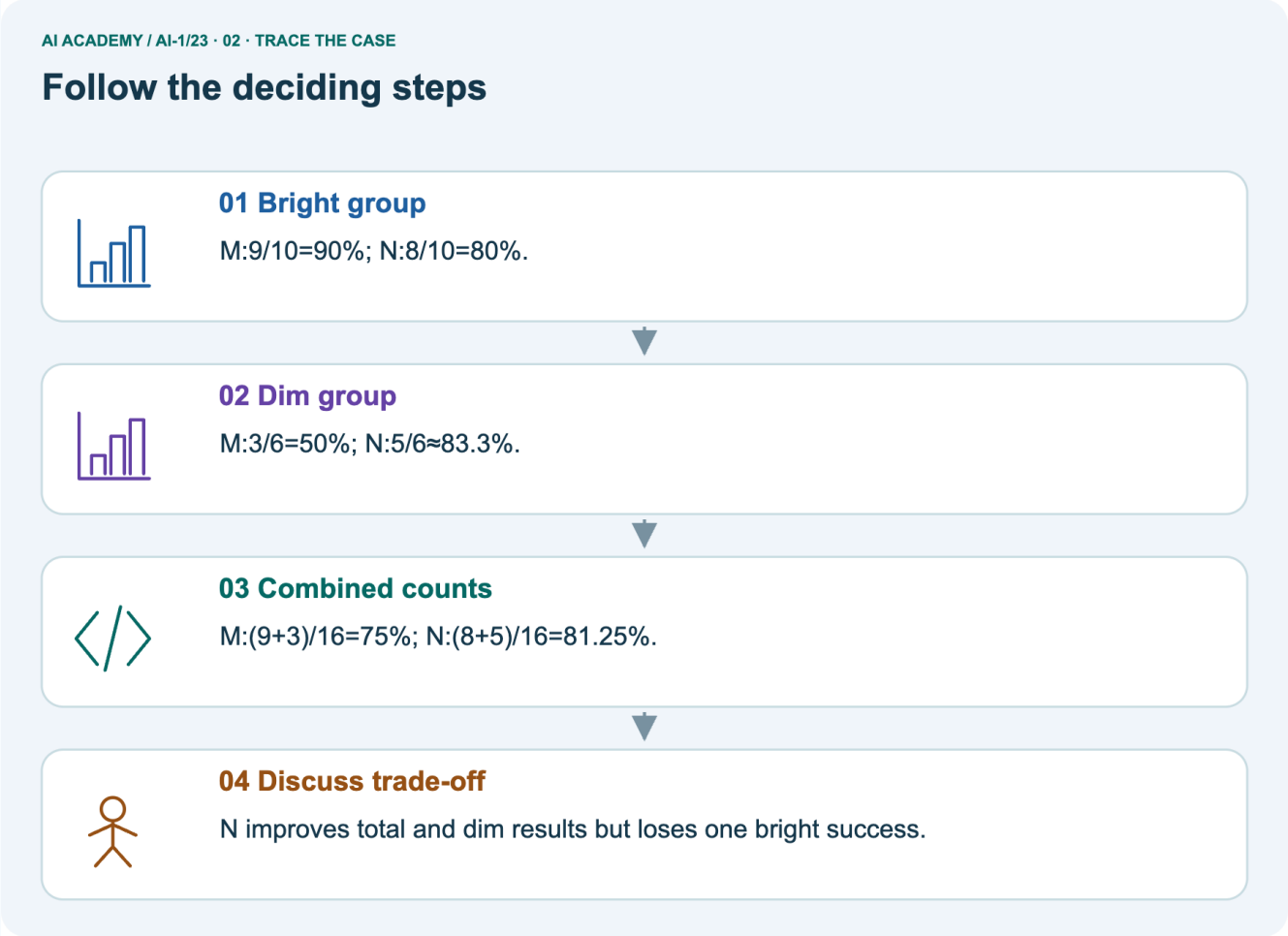
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional badge reader is evaluated on bright and dim cards with known reference labels.
Supplied facts: Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.

Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

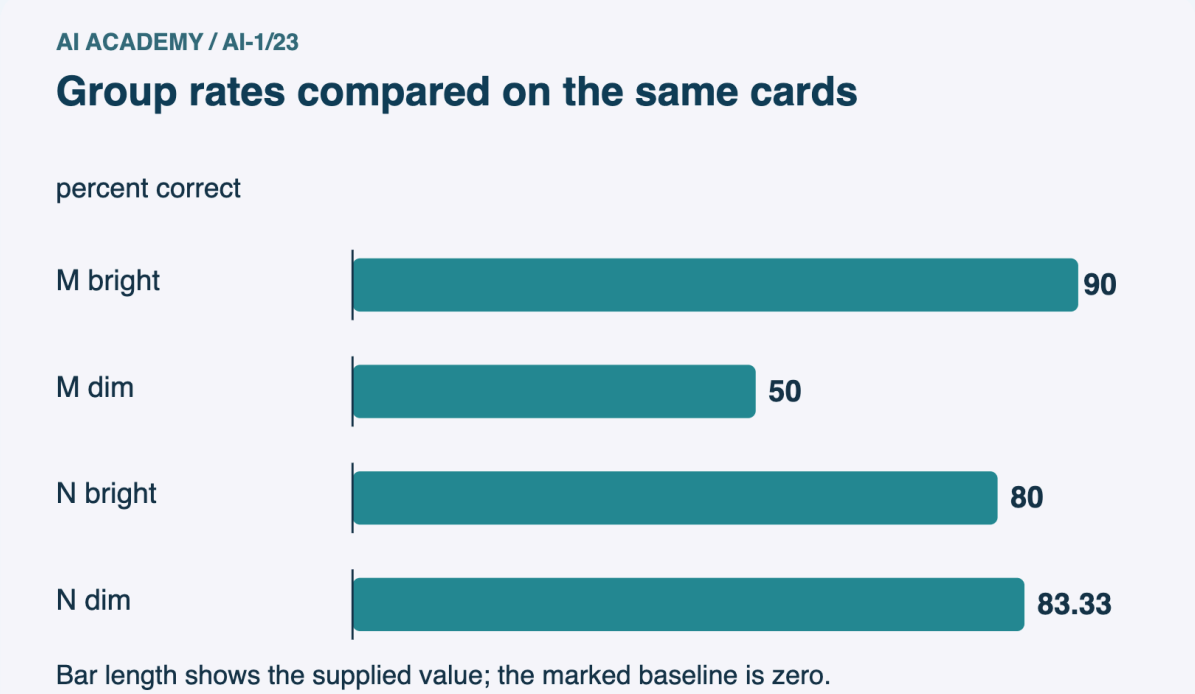


Follow the labelled picture with the matching explanation. Open the vector diagram

Bright group: M:9/10=90%; N:8/10=**80%**. → Dim group: M:3/6=50%; N:5/6≈83.3%. → Combined counts: M:(9+3)/16=75%; N:(8+5)/16=81.25%. → Discuss trade-off: N improves total and dim results but loses one bright success.

Point to each numbered stage and explain which recorded **input** it uses.

03. Group rates compared on the same cards

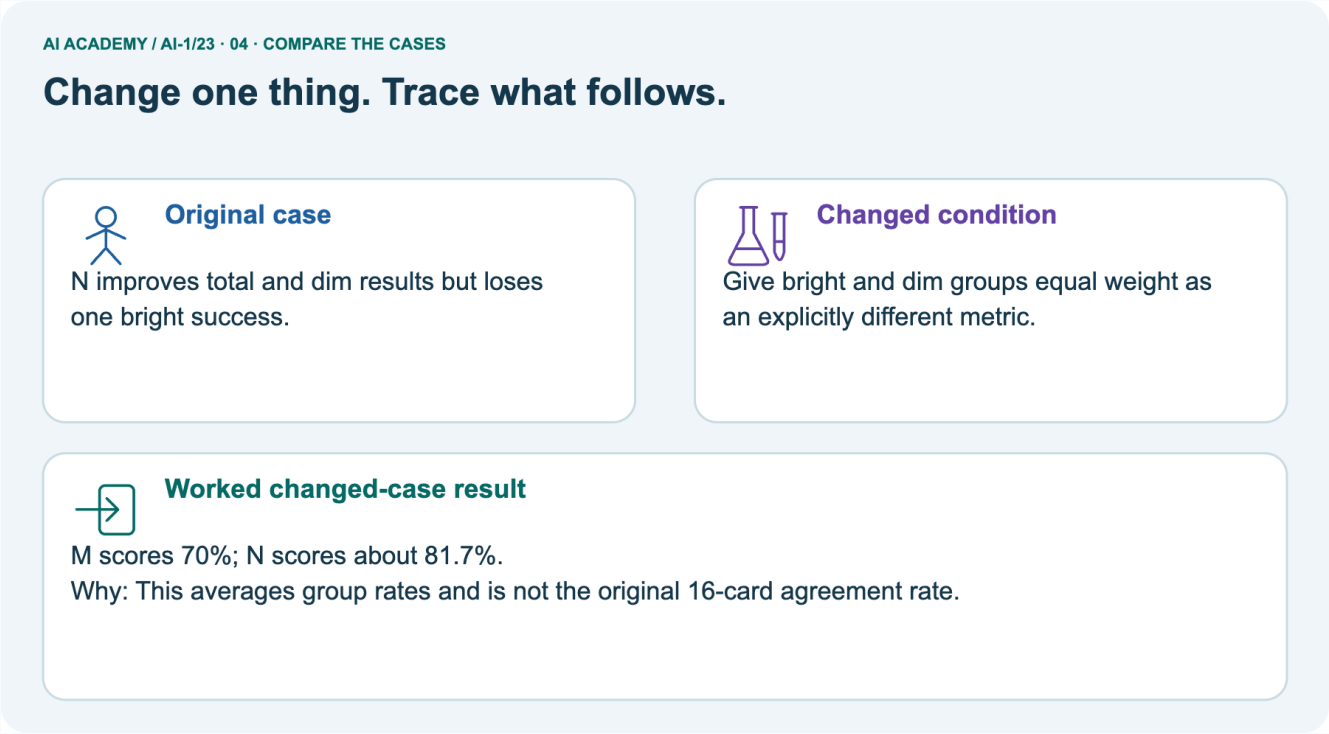


Follow the labelled picture with the matching explanation. Open the vector diagram

Group rates compared on the same cards. Units: percent correct. M bright: 90. M dim: 50. N bright: 80. N dim: 83.33333333333333.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



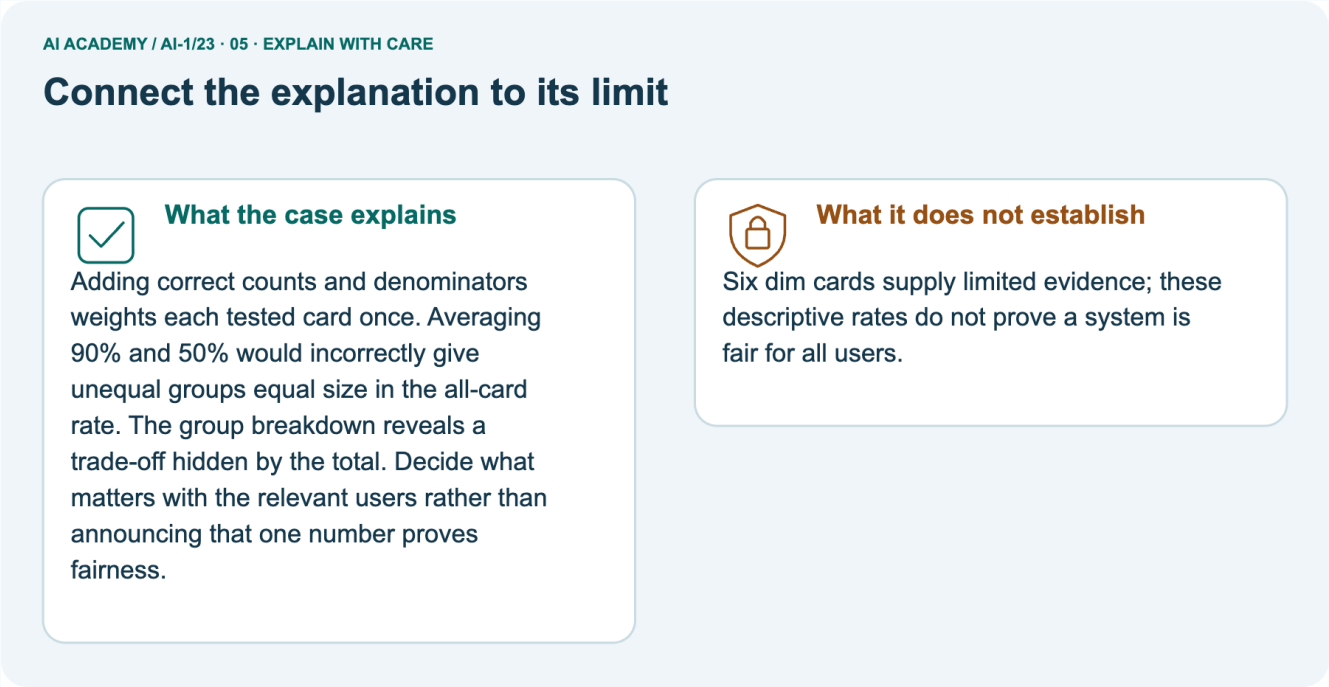
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: N improves total and dim results but loses one bright success. Changed condition: Give bright and dim groups equal weight as an explicitly different metric. Expected result: M scores **70%**; N scores about 81.7%.

Reason: This averages group rates and is not the original 16-card agreement rate.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



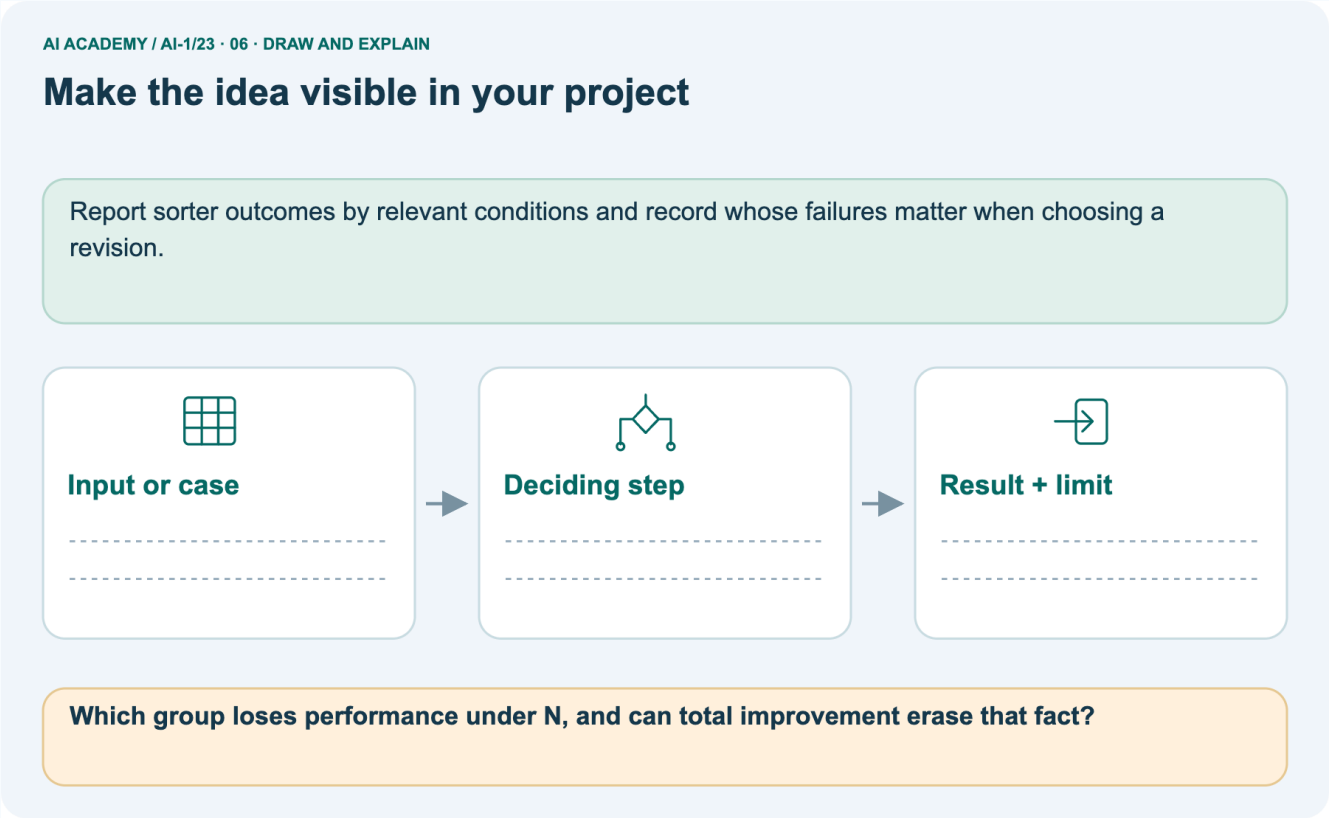
Follow the labelled picture with the matching explanation. Open the vector diagram

Adding correct counts and **denominator**s weights each tested card once. Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

Decide what matters with the relevant users rather than announcing that one number proves **fairness**. Limit: Six dim cards supply limited **evidence**; these descriptive rates do not prove a system is fair for all users.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision. Draw your **input**, deciding step, result and limit.

Supported question: Which group loses performance under N, and can total improvement erase that fact?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Fair for Whom?: the case

02. Follow the deciding steps

03. Group rates compared on the same cards

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.